NECTRA - Go-Live Checklist for Admins

Every environment variable, API key, admin setting and step needed to take NECTRA v2 from demo to production.

How to use this: work through sections 1-8 in order. Section 1 is the .env file on the server (restart the service after editing). Section 4 is Admin > Settings in the web console. Section 9 is a one-page tick list.

Tags: [REQUIRED] = must be set to go live. [RECOMMENDED] = strongly advised. [OPTIONAL] = only for that feature. [ADVANCED] = leave the default unless you know why.

1. Server environment (.env)

The file lives at the project root (nectra-platform/.env). The Lightsail installer creates it with chmod 600. After any change run:

```
sudo systemctl restart nectra
```

1.1 Core server

`DEMO_MODE=0 [REQUIRED]`

- Turns off on-screen codes, demo wallets and simulated chain actions. On every start the server applies DEMO_MODE from .env to the demo_mode setting, so this one line is enough.

`PUBLIC_URL=https://app.yourdomain.com [REQUIRED]`

- Public HTTPS address. Used in emails and as the NOWPayments IPN callback base.

`COOKIE_SECURE=1 [REQUIRED]`

- Secure cookies over HTTPS. Required for login sessions, wallets and push in production.

`TRUST_PROXY=1 [REQUIRED]`

- Set when behind nginx / Cloudflare so the real client IP is used for rate limits and audit logs.

`HOST=127.0.0.1 PORT=3000 [RECOMMENDED]`

- Bind locally; nginx proxies to it. Use HOST=0.0.0.0 only inside Docker.

`LOG_REQUESTS=0 [OPTIONAL]`

- Set 1 temporarily to log every HTTP request while debugging.

`DB_PATH=./data/nectra.db [OPTIONAL]`

- SQLite database location (WAL mode). Put it on a backed-up disk.

1.2 Secrets

`MASTER_KEY=<64 hex chars> [REQUIRED]`

- AES-256-GCM key that encrypts collector keys, live-bot wallet keys and VAPID keys. Generate with: openssl rand -hex 32. Store in a secrets manager AND an offline backup. Losing it = losing every encrypted key and the funds in live-bot wallets.

`SOLANA_COLLECTOR_SECRET=<base58 secret key> [OPTIONAL]`

- Bring your own Solana collector key (otherwise one is generated and stored encrypted).

`EVM_COLLECTOR_KEY=0x<private key> [OPTIONAL]`

- Bring your own EVM collector key (same key on all 5 EVM networks).

`TRON_COLLECTOR_KEY=<hex private key> [OPTIONAL]`

- Bring your own Tron collector key.

Collector keys only pull funds a borrower has approved. Keep them separate from the treasury, and fund them only with gas.

1.3 First-run staff accounts (only used when the database is empty)

`ADMIN_EMAIL=you@yourdomain.com [REQUIRED]`

- Super admin login email.

`ADMIN_PASSWORD=<strong password> [RECOMMENDED]`

- Super admin password. If omitted a random one is printed and saved to data/first-run-credentials.txt.

`ADMIN2_PASSWORD=<strong password> [RECOMMENDED]`

- ops@nectra.local - admin and second approver.

`OFFICER_PASSWORD=<strong password> [RECOMMENDED]`

- officer@nectra.local - loan officer.

`CREDENTIALS_FILE=<path> [OPTIONAL]`

- Where first-run credentials are written. Delete this file after you have stored the passwords.

DEMO_PASSWORD only seeds the demo trader. In production remove or disable demo@nectra.local after launch. If the database was created in demo mode, either run "npm run reset" before launch (wipes everything) or change every staff password and delete demo users.

2. Third-party APIs and accounts you need

Below is every external service NECTRA talks to, what it is for, and where to get the key.

2.1 Anthropic Claude - AI Copilot

`ANTHROPIC_API_KEY=sk-ant-... [RECOMMENDED]`

- Powers the Copilot (strategy intents, auto-tune explanations). Get it at console.anthropic.com > API Keys. Set billing limits there. Without it the rule-based parser is used.

`COPILOT_MODEL=claude-opus-5-5 [OPTIONAL]`

- Model id. Default is claude-opus-5-5.

`ANTHROPIC_AUTH_TOKEN [ADVANCED]`

- Alternative to the API key (gateway / bearer token setups).

2.2 Solana RPC - wallets, loans, treasury, live bots

`SOLANA_RPC=https://<your-provider-endpoint> [REQUIRED]`

- Dedicated mainnet RPC. The public endpoint rate-limits and will break live trading and treasury scans. Providers: Helius, Triton, QuickNode, Alchemy.

`SOLANA_CLUSTER=mainnet-beta [REQUIRED]`

- Must be mainnet-beta in production - other values switch USDC/USDT to devnet test mints (the server prints a warning at startup if it is not).

2.3 Jupiter - Solana swaps and prices

`JUPITER_API=https://lite-api.jup.ag/swap/v1 [OPTIONAL]`

-

Swap API used by live bots and the in-app Swap. For higher limits get a key/plan at portal.jup.ag and point this at the paid endpoint.

`JUPITER_PRICE_API=https://lite-api.jup.ag/price/v3 [OPTIONAL]`

- Price API for Solana tokens.

2.4 EVM RPCs - Ethereum, Base, Arbitrum, BSC, Polygon

`RPC_ETHEREUM= RPC_BASE= RPC_ARBITRUM= RPC_BSC= RPC_POLYGON= [RECOMMENDED]`

- One HTTPS RPC URL per network. Public defaults exist but are unreliable. Providers: Alchemy, Infura, QuickNode, Ankr, Chainstack. Needed for wallet verification, ERC-20 approvals, collections and repayment detection.

2.5 Tron - TronGrid

`TRON_API_KEY=<key> [RECOMMENDED]`

- Get a free key at trongrid.io (sign in > API Keys). Without it TronGrid throttles heavily.

`TRON_RPC=https://api.trongrid.io [OPTIONAL]`

- Tron full node URL.

`TRON_USDT=TR7NHqjeKQxGTCi8q8ZY4pL8otSzgjLj6t [ADVANCED]`

- TRC-20 USDT contract (mainnet default - leave as is).

2.6 Market data

`COINGECKO_API_KEY=<key> [RECOMMENDED]`

- Demo or Pro key from coingecko.com/en/api. Raises rate limits for BTC/ETH/LTC/TRX/BNB/POL prices and fallback candles.

`PYTH_API_KEY=<key> [OPTIONAL]`

- Uses Pyth Hermes for prices (the public Hermes endpoint now needs a key). Otherwise Jupiter + CoinGecko are used.

`PYTH_HERMES=https://hermes.pyth.network [ADVANCED]`

- Hermes base URL.

`CANDLES_API=https://data-api.binance.vision/api/v3/klines [OPTIONAL]`

- Hourly candles for backtests (no key needed). Falls back to CoinGecko.

`PRICE_FEED=sim [ADVANCED]`

- Forces the simulated feed. Do NOT set this in production.

2.7 BTC / LTC collateral escrow

`BTC_COLLATERAL_XPUB=<xpub/zpub> [REQUIRED for collateral loans]`

- Account-level extended PUBLIC key from a BTC wallet YOU control (e.g. hardware wallet). The server derives unique deposit addresses and can only watch, never spend.

`LTC_COLLATERAL_XPUB=<xpub/Ltub> [REQUIRED for collateral loans]`

- Same for Litecoin.

`BTC_ESPLORA=https://blockstream.info/api LTC_ESPLORA=https://litecoinspace.org/api [OPTIONAL]`

- Block-explorer APIs used to detect deposits. Self-host Esplora/mempool for reliability.

2.8 NOWPayments - crypto subscription checkout

```
NOWPAYMENTS_API_KEY=<key> [REQUIRED for crypto checkout]
```

- From account.nowpayments.io > Settings > Payments > API keys. Also set your payout wallet there.

```
NOWPAYMENTS_IPN_SECRET=<secret> [REQUIRED for crypto checkout]
```

- From the same page (IPN secret). Used to verify the HMAC-SHA512 signature on webhooks.

In the NOWPayments dashboard set the IPN callback URL to:

```
https://app.yourdomain.com/api/webhooks/nowpayments
```

2.9 Email (SMTP)

```
SMTP_HOST= SMTP_PORT=587 SMTP_USER= SMTP_PASS= [REQUIRED]
```

- Any SMTP provider: Amazon SES, Postmark, SendGrid, Mailgun, Brevo, Google Workspace. Needed for sign-up codes, 2FA/security mail, loan reminders and billing receipts. Without it mail only appears in Admin > Mailbox.

```
MAIL_FROM=NECTRA <no-reply@yourdomain.com> [REQUIRED]
```

- Sender. Verify the domain with your provider and publish SPF, DKIM and DMARC records in DNS.

2.10 Web push (VAPID)

```
VAPID_SUBJECT=mailto:you@yourdomain.com [RECOMMENDED]
```

- Contact for push services.

```
VAPID_PUBLIC_KEY= VAPID_PRIVATE_KEY= [OPTIONAL]
```

- Generate with: npx web-push generate-vapid-keys. If omitted they are auto-generated and stored encrypted - fine, but setting them keeps subscriptions valid if the database is rebuilt.

2.11 Infrastructure

- Domain + Cloudflare: A record to the server, orange cloud ON. SSL mode "Full (strict)" with a Cloudflare Origin Certificate (installer option 1).
- Server: AWS Lightsail Ubuntu, Node.js 22.5+ (24 recommended), nginx, ufw. Run: bash deploy-lightsail.sh
- Optional: lock the firewall to Cloudflare IP ranges (installer prompt).
- Squads multisig (squads.so) for the Solana treasury vault.

3. Wallets and keys you must create yourself

- Solana treasury - a Squads multisig vault address (receives repayments and crypto subscription payments; pays out loans).
- EVM treasury - a multisig (e.g. Safe) address used on Ethereum, Base, Arbitrum, BSC and Polygon.
- Tron treasury - a TronLink / hardware-wallet address (multisig permissions recommended).
- BTC and LTC hardware wallets - export the account xpub for section 2.7.
- Collector gas - after launch open Admin > Treasury & keys and send a small amount of SOL, ETH, BNB, POL, ETH-on-Base/Arbitrum and TRX to each collector address so collections can pay fees.
- MASTER_KEY backup - printed/offline copy kept apart from the server.

4. Admin console settings (Admin > Settings)

Sign in at https://app.yourdomain.com/admin as the super admin. Only super admins can change settings.

4.1 Platform

- demo_mode -> OFF.
- registration_open -> on/off as you want sign-ups.
- invite_required -> decide (default on). Replace the demo invite code NECTRA26 with your own codes.
- maintenance_banner -> empty (use it for planned downtime).
- paper_start_usdc / paper_start_sol -> starting paper balances for new users (default 10000 / 20).
- withdraw_hold_hours -> hold after security changes before withdrawals (default 24).
- referral_max_codes / referral_code_uses -> referral limits (default 5 / 5).

4.2 Security and approvals

- admin_2fa_required -> ON. First enable authenticator 2FA on your own account (Settings > Security), then every staff account.
- dual_approval_threshold -> USD amount at/above which loans, payouts and balance adjustments need a second admin (default 1000). Every collection always needs two admins.
- Roles -> give each staff member the least role needed (loan officer < admin < super admin). Keep at least two admins so approvals are possible.

4.3 Treasury

- treasury_solana -> Squads vault address.
- treasury_evm -> EVM treasury address.
- treasury_tron -> Tron treasury address.

Repayments and on-chain USDC subscription payments are only detected once these are set.

4.4 Lending

- default_after_days -> days past due before a loan is marked defaulted (default 30).
- Loan products (Admin > Loans / Products) -> set amounts, APR, instalments, late-fee %, grace period and the collateral product (150% BTC/LTC).
- Review the loan book and CSV export work before opening applications.

4.5 Trading

- live_trading_enabled -> ON only when SOLANA_RPC and MASTER_KEY are production-ready (default off).
- live_bot_max_usd -> per-bot funding cap (default 1000).
- ai_copilot -> ON to use Claude when ANTHROPIC_API_KEY is set.

4.6 Monetization

- monetization_enabled -> master switch for plans and gating (default on).
- crypto_payments_enabled -> allow NOWPayments / on-chain USDC checkout (default on).
- nowpayments_enabled -> ON after NOWPAYMENTS_API_KEY and IPN secret are set (default off).
- Admin > Plans & subscriptions -> review Free / Starter $19 / Pro $49 / Elite $149, annual

prices and perks.

4.7 Compliance

- current_terms_version -> set after your lawyer approves the Terms & Risk Disclosure copy; bump it to force re-acceptance.
- kyc_enabled -> master switch for identity verification (default on).
- kyc_required_for_loans -> default on.
- kyc_required_for_live -> default off; decide with counsel.
- Admin > KYC review -> assign reviewers and test approve/reject.

5. Deployment steps (Lightsail)

1. Create the Lightsail Ubuntu instance; open ports 80/443 only.
2. git clone the repo, cd into it, run: bash deploy-lightsail.sh. Answer: your domain, demo mode = n, SSL option 1 (paste Cloudflare Origin Certificate), firewall lock = y.
3. Edit .env with every value from sections 1 and 2, then: sudo systemctl restart nectra.
4. Check logs: sudo journalctl -u nectra -n 100 --no-pager. The startup banner must say Demo mode: off.
5. In Cloudflare: DNS A record (proxied), SSL/TLS = Full (strict), Always Use HTTPS = on.
6. Open https://app.yourdomain.com/status and confirm prices, RPCs and mail are healthy.
7. Sign in to /admin, store the first-run passwords, delete data/first-run-credentials.txt, enable 2FA.
8. Complete the Admin > Settings values in section 4.

6. Backups and operations

- Back up data/nectra.db (plus -wal and -shm, or use sqlite3 .backup) daily to off-server storage.
- Back up .env and MASTER_KEY separately and securely.
- Keep the OS patched; restrict SSH to your IP.
- Watch Admin notifications for risk flags (revoked approvals, wallet value < 25% of debt) and unmatched deposits.
- Rotate staff passwords and API keys periodically; remove staff who leave.

7. Pre-launch tests

1. Run npm test on the server copy (50 end-to-end checks on a throwaway DB).
2. Register a real user: email code arrives by SMTP.
3. Connect Phantom, MetaMask and TronLink; balances load.
4. Push notification arrives on desktop and on an iPhone Home Screen install.
5. Subscribe with a small NOWPayments payment; the IPN activates the plan.
6. Send a small USDC repayment to the treasury; it is detected and applied.
7. Create a small live bot, fund it, confirm a real Jupiter swap, then withdraw.
8. Request a collection with one admin and approve with a different admin.
9. Copilot answers using Claude (not the rule-based fallback).

8. Legal and compliance

- Get legal advice for lending, holding account balances, collecting from customer wallets, custodial bot wallets and KYC/AML in every jurisdiction you serve.
- Finalize Terms, Risk Disclosure and Privacy Policy before setting current_terms_version.
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Decide whether manual KYC is enough or wire a KYC provider (e.g. Sumsub, Persona, Onfido).
- Plans gate features only - never market them as increasing profit.

9. One-page tick list

- [ ] DEMO_MODE=0 (startup banner shows Demo mode: off)
- [ ] MASTER_KEY set and backed up offline
- [ ] PUBLIC_URL, COOKIE_SECURE=1, TRUST_PROXY=1
- [ ] Strong ADMIN / ADMIN2 / OFFICER passwords; credentials file deleted
- [ ] ANTHROPIC_API_KEY
- [ ] SOLANA_RPC (dedicated) + SOLANA_CLUSTER=mainnet-beta
- [ ] RPC_ETHEREUM, RPC_BASE, RPC_ARBITRUM, RPC_BSC, RPC_POLYGON
- [ ] TRON_API_KEY
- [ ] COINGECKO_API_KEY (PYTH_API_KEY optional)
- [ ] BTC_COLLATERAL_XPUB, LTC_COLLATERAL_XPUB
- [ ] NOWPAYMENTS_API_KEY, NOWPAYMENTS_IPN_SECRET, IPN URL set in dashboard
- [ ] SMTP_* + MAIL_FROM, SPF/DKIM/DMARC
- [ ] VAPID_SUBJECT (+ VAPID_PUBLIC_KEY / VAPID_PRIVATE_KEY)
- [ ] Cloudflare DNS + Full (strict) origin certificate
- [ ] Treasury addresses: Solana (Squads), EVM, Tron
- [ ] Collector addresses funded with gas
- [ ] admin_2fa_required ON, dual_approval_threshold set, roles reviewed
- [ ] Loan products, default_after_days reviewed
- [ ] live_trading_enabled + live_bot_max_usd decided
- [ ] Monetization, crypto, NOWPayments toggles set; plans reviewed
- [ ] Terms version, KYC switches decided
- [ ] Daily DB backups running
- [ ] Pre-launch tests in section 7 all pass
- [ ] Legal sign-off